SIGNATURE VERIFICATION SYSTEM

Board of Directors - Investment & Approval Proposal

AI-Powered Fraud Prevention for Banking Operations

Prepared by: **Business Analyst + AI Team** | Version 1.0 | Confidential

OVERVIEW

What Is This System and How Does It Work?

WHAT IT IS

An AI-powered tool that checks whether a signature on a document belongs to the person it claims to be from.

A customer's genuine signature is stored once when they open an account. Every time a signed document arrives — a cheque, a form, an agreement — the system compares it against that stored reference and returns a clear verdict.

The result: MATCH or NO MATCH, with a confidence score and a full audit trail.

HOW IT WORKS — STEP BY STEP

1. Document with signature arrives at the bank
2. Signature image is scanned and uploaded to the system
3. System cleans and standardises the image automatically
4. AI converts the signature into a unique numerical pattern
5. Pattern compared against the customer's stored reference
6. MATCH or NO MATCH returned in under 90-120 milliseconds

APPLICATIONS & OUTCOMES

Where It Applies and What We Can Achieve

Use cases across banking operations and measurable outcomes

BANKING USE CASES

Use Case	How the System Helps
Cheque Processing	Flags signatures that do not match account holder records before clearing
Loan & Mortgage Documents	Confirms signatory identity on high-value agreements
Account Opening / Closure	Verifies that forms are signed by the actual account holder
Internal Approvals	Checks that authorised signatories have signed payroll and expense forms
Proof of Delivery	Matches recipient signature against enrolled customer profile

ACHIEVABLE OUTCOMES

92–95%
Verification Accuracy

<90ms
Response Time per verify

~60%
Reduction in Manual Work

7 yrs
Audit Log Retention

Important: The system is a decision-support tool. It reduces manual review workload — it does not replace human officer judgement for high-value or disputed transactions.

QUALITY & CONFIGURATION

Image Quality Requirements and Threshold Settings

What is needed for the system to work accurately?

IMAGE QUALITY STANDARDS

Property	Required	Why It Matters
Resolution	300 DPI minimum	Fine stroke detail is preserved
Background	Plain white paper	No lines or colour confusing the AI
Ink	Black or dark blue pen	Maximum contrast against background
Format	PNG file	No image compression that loses detail
Lighting	Even, no shadows	Shadow creates false ink patterns
Orientation	Not tilted > 5°	Excessive rotation distorts matching

RISK-BASED THRESHOLD SETTINGS

Transaction Type	Score	Action
Low value cheque < NPR 50,000	≥ 0.80	Auto-approve
Standard cheque NPR 50K–500K	≥ 0.85	Auto-approve
High value cheque > NPR 500,000	≥ 0.92	Officer review
Loan documents	≥ 0.95	Always officer
Any NO MATCH	< min	Flag for review

CONFIDENCE SCORE GUIDE

0.95 – 1.0 Very High	0.90 – 0.94 High	0.85 – 0.89 Medium	0.70 – 0.84 Low	< 0.70 Very Low

TRAINING & LIMITATIONS

Training Cycle and Known Limitations

What is needed once to make it work — and where human oversight is still required

HOW TRAINING WORKS

ONCE	Collect 500+ signature samples — genuine and imitation — from a sample of customers
ONCE	Run training on dedicated hardware (~1 hour with GPU)
ONCE	Deploy the trained model to the production API server
ONGOING	New customers enrol by uploading one reference signature — no retraining needed
QUARTERLY	Collect misclassified cases from officer review queue and retrain to improve

Train once. Enrol customers in seconds. Improve quarterly.

KNOWN LIMITATIONS

Accuracy 92–95%, not 100%
High-value transactions always go to officer review

Requires enrolled reference first
Enrolment must happen at account opening

Image quality affects results
Poor scans produce unreliable scores

Skilled forgeries can fool the system
Used alongside other checks, not standalone

Signatures change over time
Re-enrolment needed every 1–2 years per customer

Phased Deployment Plan and Resource Requirements

What we are asking the Board to approve — and what will be delivered

THREE-PHASE DEPLOYMENT

01 Prototype & Pilot

Months 1 – 2

- Deploy at 1 branch
- Enrol 500 customers
- Run alongside manual checking
- Validate accuracy on real data

02 Assisted Mode

Months 3 – 4

- System shows recommendation
- Officer makes final decision
- Tune thresholds per transaction
- Collect misclassified cases

03 Semi-Automated

Month 5 onwards

- Low-value: auto-approve on high confidence
- High-value: always officer review
- Retrain model quarterly
- Expand to all branches

2 Developers · 6 weeks — Pilot build | 1 Data Collector · 3 months — Signatures | 1 GPU Server · Training + inference | 1 Officer · Review queue oversight